12. Introducing Turbots

Embodied Computation

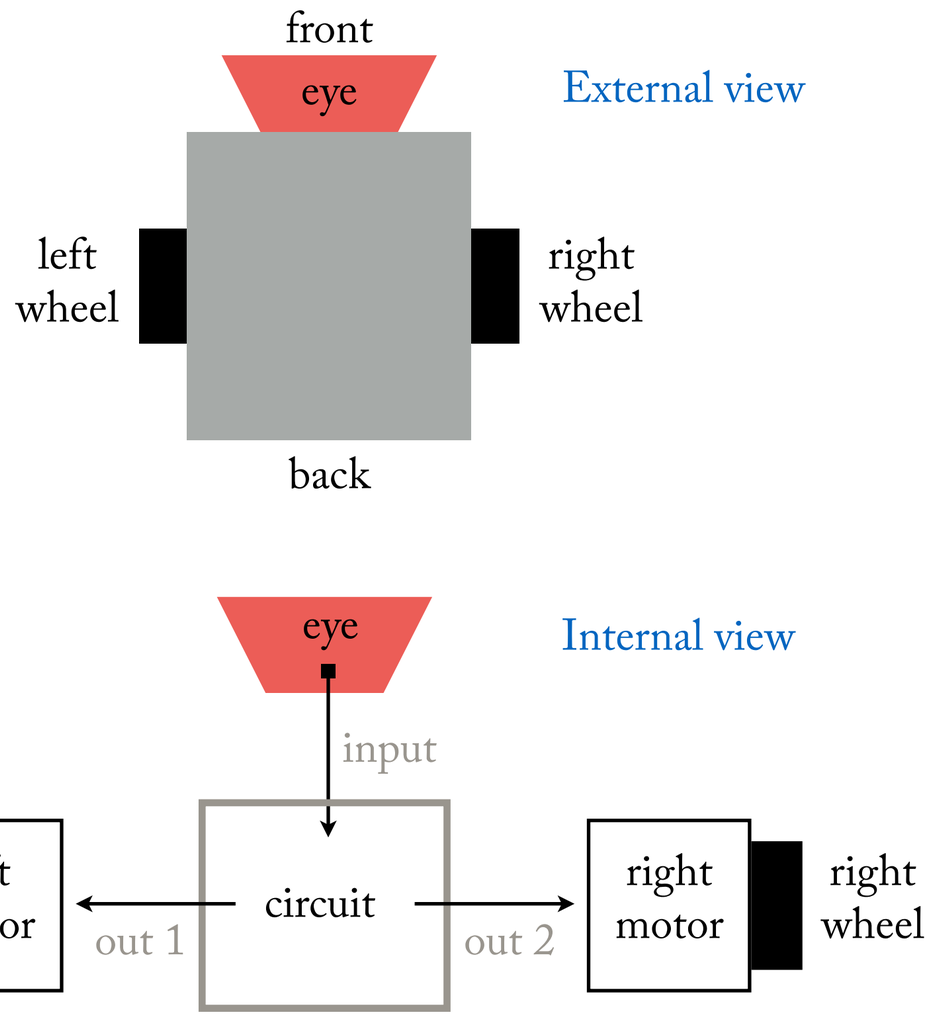
The central fact about cognition is that it allows an organism to interact meaningfully with its environment. Cognition mediates between an organism's ability to sense the environment and its ability to act on it. We can't claim to be studying cognition by only focusing on mathematical computation, even if that is an important part of the story. In fact, many believe that for a computer to exhibit *genuine* mental representation, it must be part of a fully embodied organism that causally interacts with the physical world.

For these reasons, we will now begin to investigate computational problems of navigation for organisms capable of perception and action, alongside the more abstract, traditional study of mathematical computation.

Turbots

A **turbot** (from "Turing" + "robot") is a simple robot with a single electronic eye, two wheels powered by two motors, and an internal circuit which processes signals from the eye, and issues signals to the two motors.

For perception, turbots are limited to seeing the object immediately in front of them; for action, they can only turn left or right, go straight, or stop.

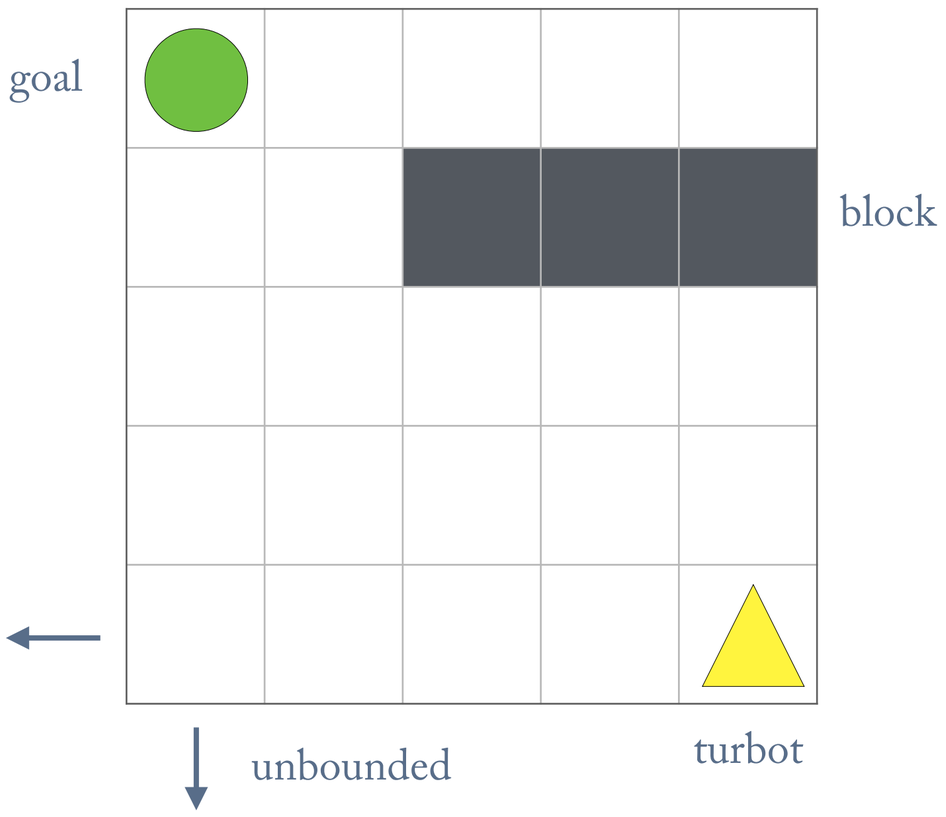


Environment

Turbots live in an unbounded grid-world that includes the turbot, unmovable blocks, and (typically) a goal.

The position and orientation of the turbot are indicated by the yellow triangle.

For navigation problems, the yellow triangle indicates the turbot's starting point. Typically, navigation is "completed," and the problem solved, when the turbot *crosses over* the goal. It does not need to stop on the goal. (Think of Pac-Man.)



Progression of Time

During each unit of time, or **cycle**, the turbot "instantly":
(1) receives a new input;
(2) processes it;
(3) produces an output;
(4) acts on that output.

At the end of each cycle the circuit is "wiped clean" of any information, but the circuit itself persists through time, and the process begins again. (By contrast, for CCs that compute mathematical functions, we only consider their operation at a single moment in time, for a single input.)

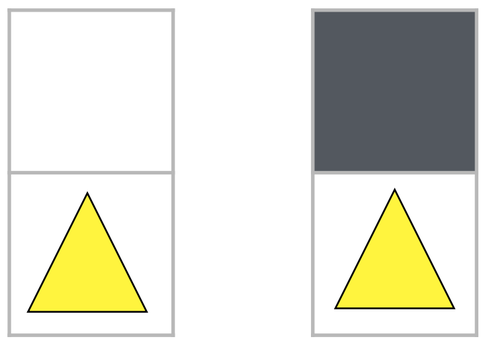
Input: Perception

The input to the turbot comes through the eye. At each cycle, the eye can only see the cell in front of the turbot at that time. And it can only detect the absence (input = 0) or presence (input = 1) of a block.

For now, turbots cannot see goals, only empty cells and blocks.

Turbot perception

empty → 0
block → 1



empty
input = 0

block
input = 1

Output: Action

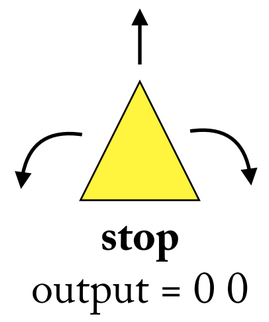
The output from the turbot goes through the two motors and on to the wheels. An output of 1 to a motor will turn it on for 1 cycle, while an output of 0 will turn it off for 1 cycle.

If both motors receive a 1, the turbot moves forward 1 cell. If both motors receive a 0, the turbot stops in the current cell.

If the *left* motor receives a 1 and the *right* motor receives a 0, the turbot will rotate 90° *to the right*, but it will not change cells. (This makes mechanical sense if you think about it.) The opposite is also true: if the *right* motor receives a 1 and the *left* motor a 0, the turbot as a whole will rotate 90° *to the left*.

The only way to turn 180° is to apply two 90° turns over two cycles.

forward 1 cell
output = 1 1



90° left
output = 0 1

90° right
output = 1 0

stop
output = 0 0

Turbot action

1 1 → forward 1 cell
0 1 → 90° left
1 0 → 90° right
0 0 → stop

A Simple Example

Consider the following problem:

> Design a turbot which moves straight ahead until it reaches a block then stops.

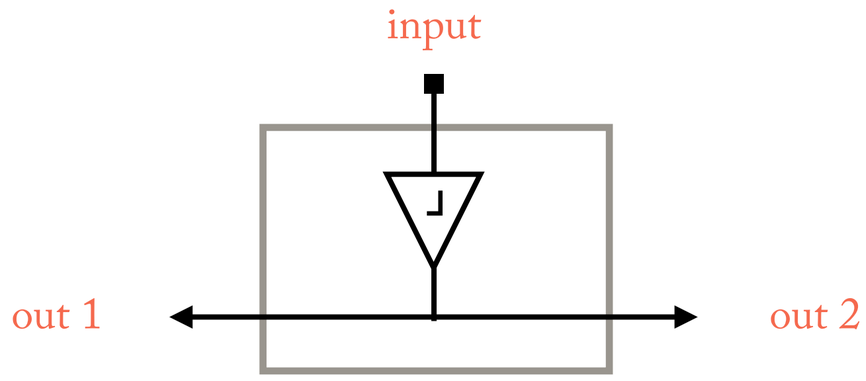
To satisfy this description, the turbot must repeatedly apply the following rule:

> If perception = **empty**, then action = **move forward**.
> If perception = **block**, then action = **stop**.

In terms of inputs and outputs:

> If IN = **0**, then OUT = **1 1**.
> If IN = **1**, then OUT = **0 0**.

It is now trivial to design such a circuit:



Installed into a turbot, this circuit solves the problem we started with.

